Experiment :- 20

Problem statement:-

Mathematical computations such as finding the Greatest Common Divisor (GCD), Least Common Multiple (LCM), and checking whether a number is prime are fundamental problems in programming and algorithm design. Automating these computations using shell scripting helps students understand arithmetic operations, conditional logic, and iterative constructs in Bash.

The problem is to write a shell script that accepts user input to compute the GCD and LCM of two numbers and checks whether a given number is prime, demonstrating arithmetic operations, loops, and decision-making in Bash scripting.

Aim:- Shell script to compute GCD. LCM and check whether a given number is prime.

Procedure:-nano

A. Writing the Shell Script

1. Open the terminal.

2. Create a script file: nano math_ops.sh

```
ubuntu@ubuntu:~$ nano maths_ops.sh
```

3. Enter the script:

```
File  Actions  Edit  View  Help
  GNU nano 8.4                                                    maths_ops
#!/bin/bash
echo "enter first number:"
read a
echo "enter second number:"
read b
x=$a
y=$b
# GCD calculation
while [ $y -ne 0 ]
do
   temp=$y
   y=$((x % y))
   x=$temp
done
gcd=$x
1cm=$(( (a*b) / gcd ))
echo "GCD of $a and $b is: $gcd"
echo "LCM of $a and $b is: $1cm"
echo "enter a number to check for prime :"
read n
flag=0
if [ $n -le 1 ];then
   flag=1
fi
for ((i=2; i<=n/2; i++))
do
   if [ $ ((n % i )) -eq 0 ]; then
      flag=1
      break
   fi
```

```
  GNU nano 8.4
# GCD calculation
while [ $y -ne 0 ]
do
   temp=$y
   y=$((x % y))
   x=$temp
done
gcd=$x
1cm=$(( (a*b) / gcd ))
echo "GCD of $a and $b is: $gcd"
echo "LCM of $a and $b is: $1cm"
echo "enter a number to check for prime :"
read n
flag=0
if [ $n -le 1 ];then
   flag=1
fi
for ((i=2; i<=n/2; i++))
do
   if [ $ ((n % i )) -eq 0 ]; then
      flag=1
      break
   fi
done
if [ $flag -eq o ]; then
   echo "$n is a prime number"
else
   echo "$n is not a prime number"
fi
```

4. Save and exit the editor.

5. Make the script executable: chmod +x math_ops.sh

```
ubuntu@ubuntu:~$ nano maths_ops.sh
ubuntu@ubuntu:~$ chmod +x maths_ops.sh
```

B. Executing the Script 6. Run the script:

7. ./math_ops.sh

```
└$ ./maths_ops.sh
enter first number:
5
enter second number:
3
./maths_ops.sh: line 16: 1cm=15: command not found
GCD of 5 and 3 is: 1
LCM of 5 and 3 is: cm
enter a number to check for prime :
```